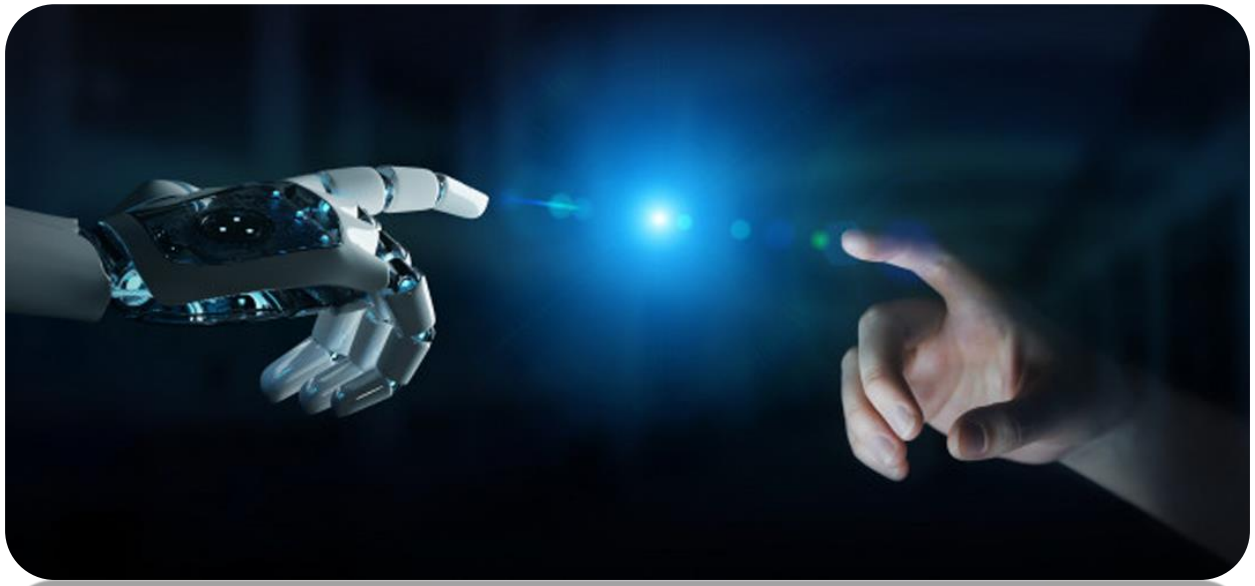


Robotics

Contents



Chapter 1. Representing Position and Orientation



Chapter 2. Image Formation



Chapter 3. Control



Chapter 4. Localization and Mapping



Chapter 5. Robot Arm Kinematics

DRAFT

Grading



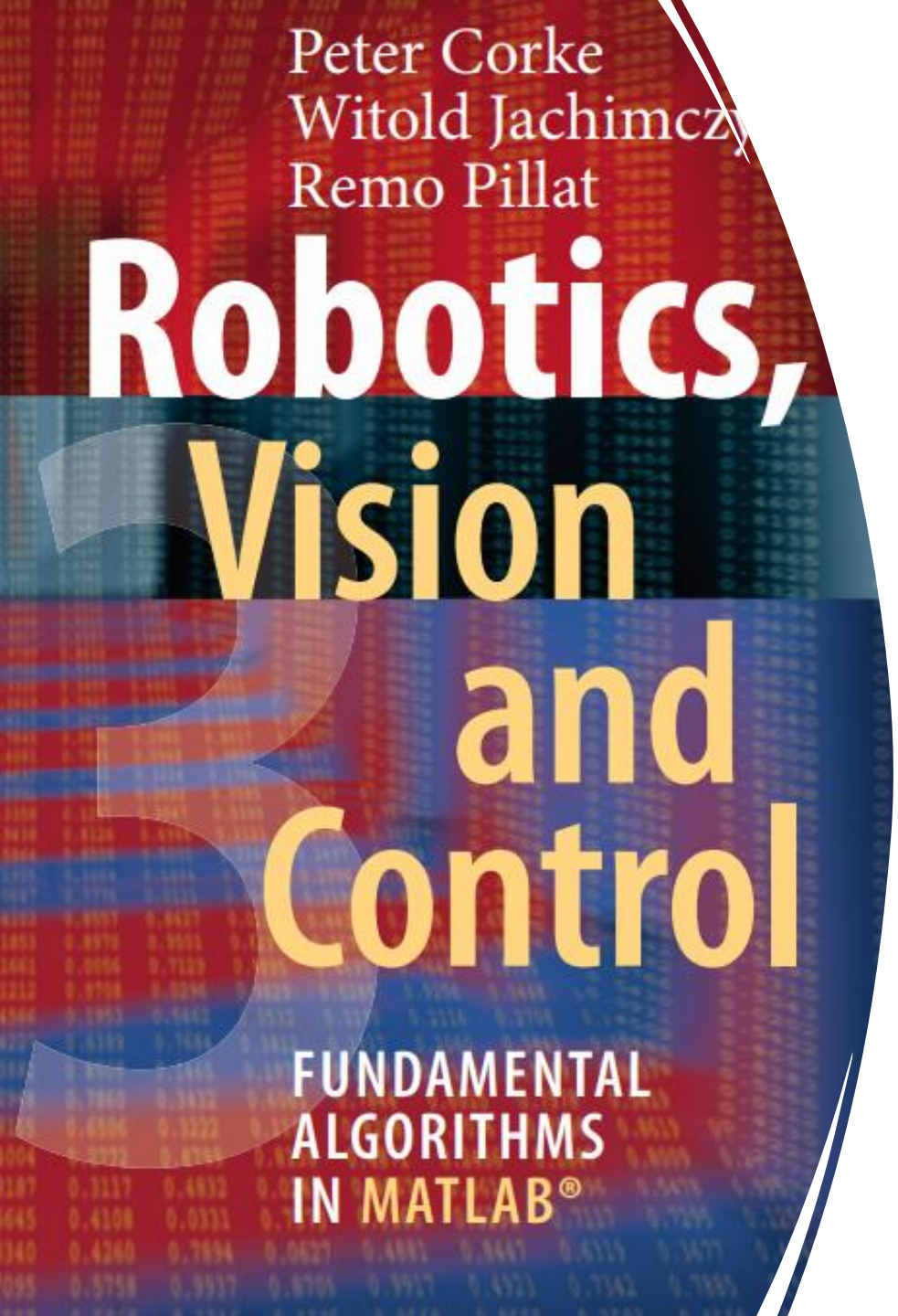
Homeworks: 20%



Mid-term (project): 30%



Final term (project): 50%



Corke, Peter I., Witold Jachimczyk, and Remo Pillat. ***Robotics, vision and control: fundamental algorithms in MATLAB***. 3rd edition.

- *Tell me and I will forget.*
  Show me and I will remember.
  Involve me and I will understand.
  – Chinese proverb

- *Simple things should be simple,*
  complex things should be possible.
  – Alan Kay

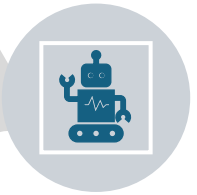
About the course



the software tools used in this course aim to reduce complexity for the learner.



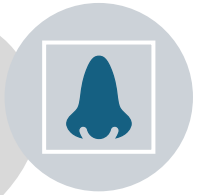
allow the learner to work with real problems, not just trivial examples.



a cohesive narrative that covers robotics and computer vision – both separately and together.



show how complex problems can be decomposed and solved.



consider the course as a grand tasting menu.



software is a first-class citizen in this course.



instant gratification in just a couple of lines of MATLAB code.



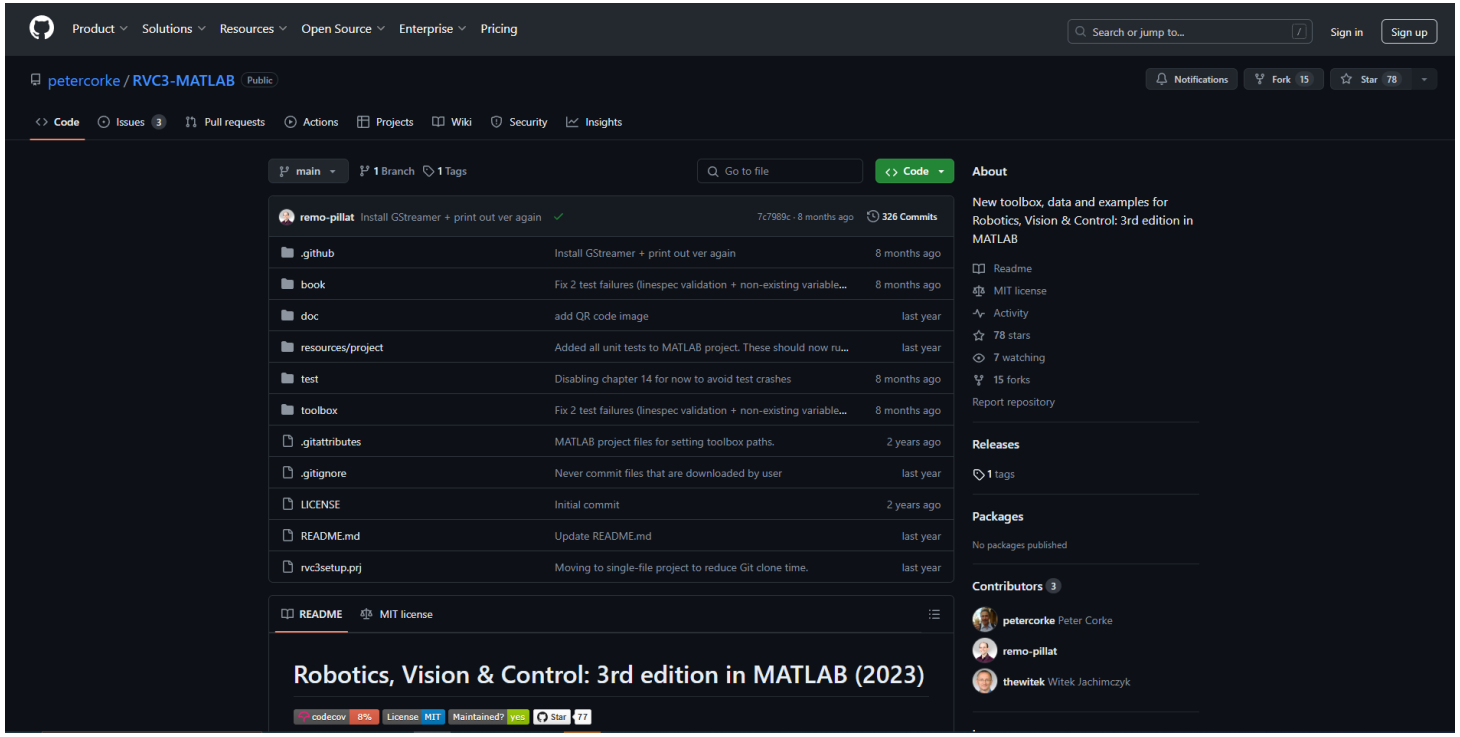
this course provides a complementary approach.

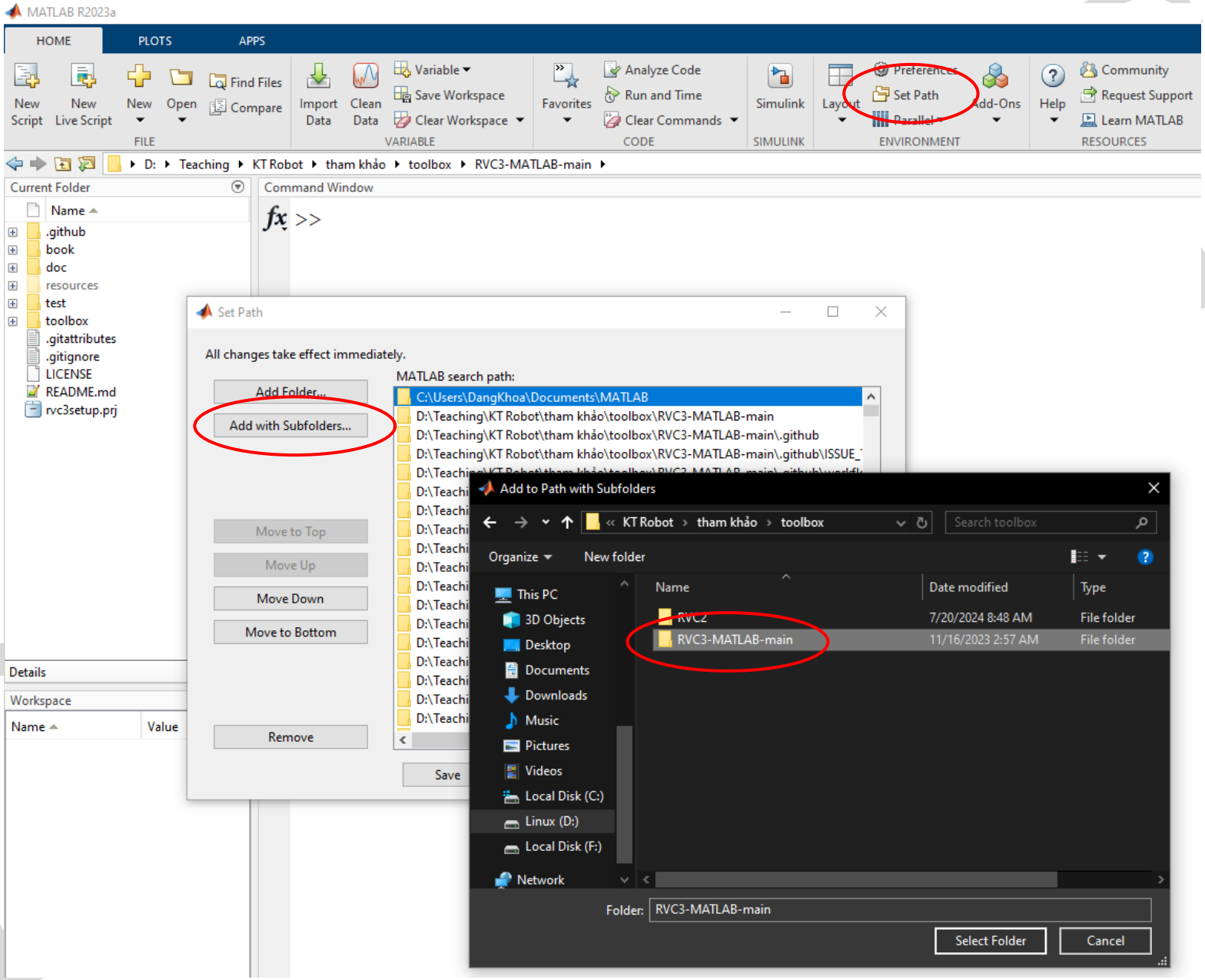
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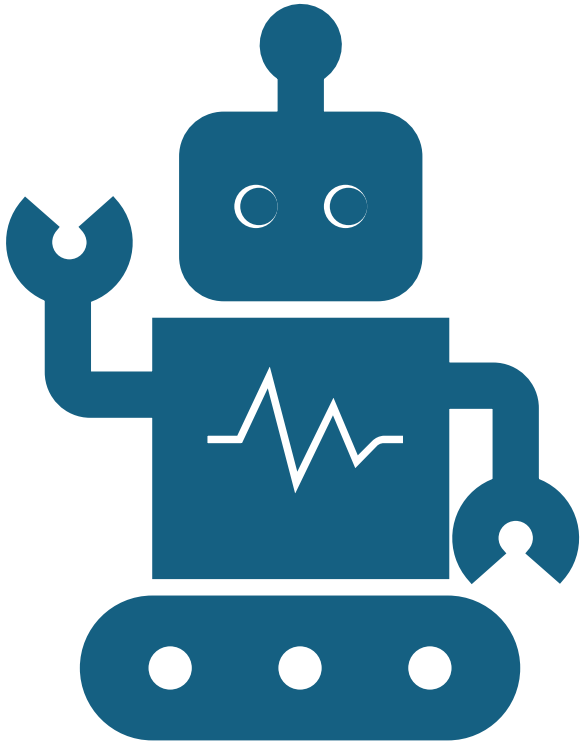
Toolbox RVC3-MatLab

- Github:

https://github.com/petercorke/RVC3-MATLAB



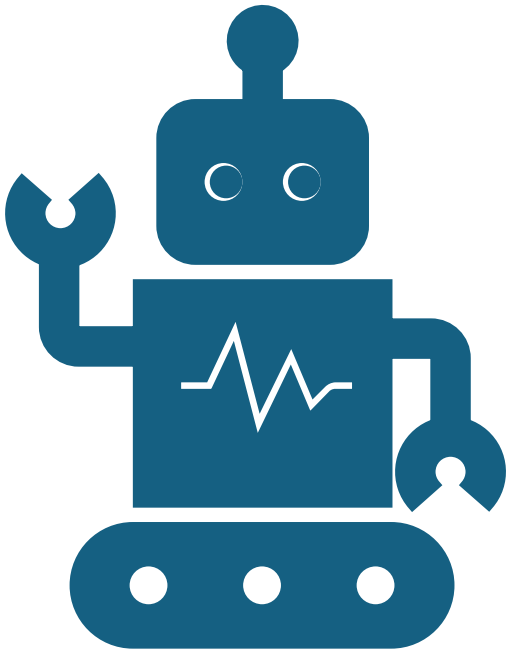

MATLAB R2023a
HOME
PLOTS
APPS
New Script
New Live Script
New
Open
Find Files
Compare
Import Data
Clean Data
Variable
Save Workspace
Clear Workspace
Favorites
Analyze Code
Run and Time
Clear Commands
Simulink
Layout
Preferences
Set Path
Parallel
Add-Ons
Help
Community
Request Support
Learn MATLAB
FILE
VARIABLE
CODE
SIMULINK
ENVIRONMENT
RESOURCES
D: ▸ Teaching ▸ KT Robot ▸ tham khảo ▸ toolbox ▸ RVC3-MATLAB-main ▸
Current Folder
Name
.github
book
doc
resources
test
toolbox
.gitattributes
.gitignore
LICENSE
README.md
rvc3setup.prj
Command Window
>>
Details
Workspace
Name
Value
Set Path
All changes take effect immediately.
Add Folder...
Add with Subfolders...
Move to Top
Move Up
Move Down
Move to Bottom
Remove
MATLAB search path:
C:\Users\DangKhoa\Documents\MATLAB
D:\Teaching\KT Robot\tham khảo\toolbox\RVC3-MATLAB-main
D:\Teaching\KT Robot\tham khảo\toolbox\RVC3-MATLAB-main\.github
D:\Teaching\KT Robot\tham khảo\toolbox\RVC3-MATLAB-main\.github\ISSUE_
Save
Add to Path with Subfolders
« KT Robot › tham khảo › toolbox
Search toolbox
Organize
New folder
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
Linux (D:)
Local Disk (F:)
Network
Name
Date modified
Type
RVC2
7/20/2024 8:48 AM
File folder
RVC3-MATLAB-main
11/16/2023 2:57 AM
File folder
Folder:
RVC3-MATLAB-main
Select Folder
Cancel

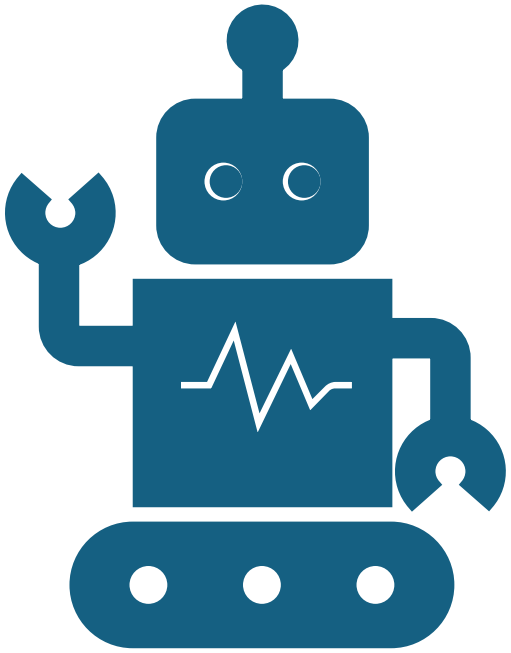


What is a robot?

I can't define a robot, but I know one when I see one.

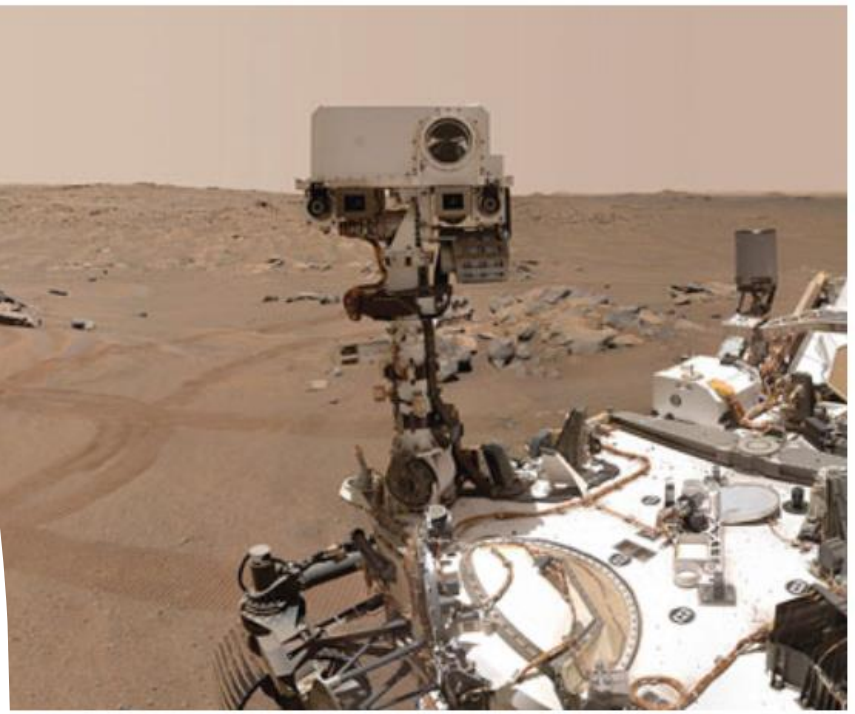
– Joseph Engelberger





a goal-oriented machine that can sense, plan, and act.

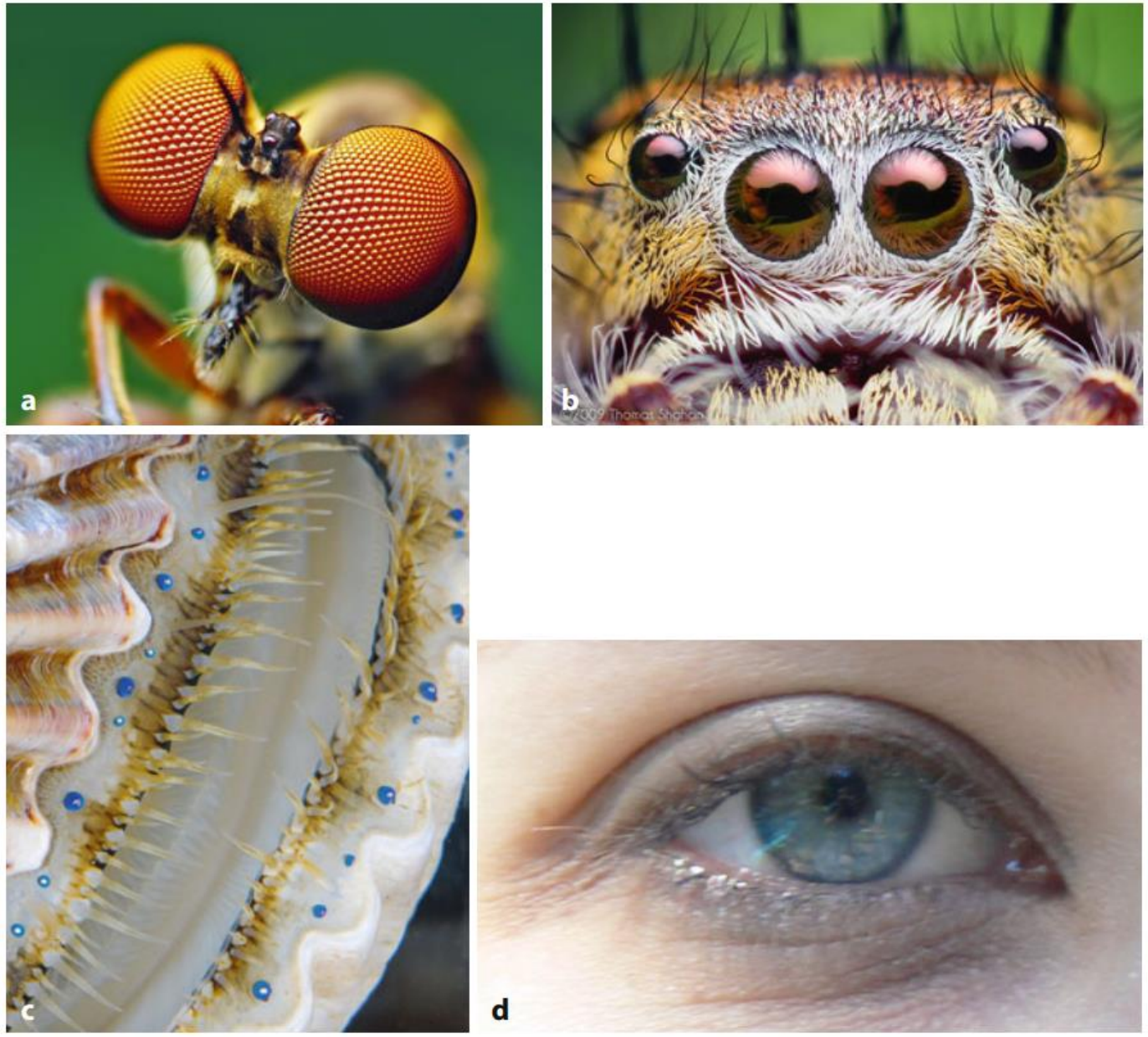








Robotic Vision



Computer Vision





Ethical Considerations